

Transport coefficients of dense matter in the liquid metal regime*

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Summary. We have calculated the thermal and electrical conductivities and the shear viscosity of dense matter when the ions form a liquid. Effects of ionic correlations are included by use of results for the liquid structure factor of the one component plasma obtained from numerical simulations, and screening of the electron–ion interaction is taken into account. At densities so high that screening is unimportant our results for the thermal and electrical conductivities agree to within ~ 10 per cent with earlier ones by Yakovlev & Urpin. We also estimate electron effective mass and vertex corrections.

1 Introduction

Transport coefficients of dense matter are an important ingredient in studies of neutron star evolution. For example, it has recently been shown (Gudmundsson, Pethick & Epstein 1982) that the thermal conductivity of matter in the regime where the ions form a liquid metal is particularly important for calculations of neutron star cooling. Transport coefficients of dense matter were first calculated by Marshak (1941), and since then a number of calculations have been made (Mestel 1950; Lee 1960; Hubbard & Lampe 1969; Ewart, Guyer & Greenstein 1975; Flowers & Itoh 1976, FI; Urpin & Yakovlev 1980; Yakovlev & Urpin 1980, YU).

At densities where there are discrete nuclei, there are significant differences between the YU and FI calculations. In the case of the thermal conductivity, the FI results are typically 3–5 times the YU ones. Raikh & Yakovlev (1982) have carried out more detailed calculations of the thermal conductivity when the ions are solid, and find results that agree well with the YU ones. In this paper we calculate transport coefficients of dense matter when the ions form a liquid.

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2 Basic Formalism

Under the conditions of interest, all transport is chiefly by electrons, and the electron mean free path is limited by collision of electrons with ions. The electron–electron interaction energy is of order $e^2/\hbar v_F$ times the electronic kinetic energy, where v_F is the electron Fermi velocity, and therefore provided $v_e/c \gg e^2/\hbar c$, or $\rho \gg 10^4 \text{ g cm}^{-3}$, the electrons may be treated as free. Expressions for transport coefficients of a degenerate electron gas have been derived by Baym (1964) for the non-relativistic case, and by FI for the relativistic case. One finds for the electrical conductivity, σ , the thermal conductivity, κ , and the shear viscosity, η ,

$$\sigma = \frac{n_e e^2 \tau_\sigma}{m_R} \quad (1)$$

$$\kappa = \frac{\pi^2}{3} n_e \frac{k_B^2 T}{m_R} \tau_\kappa \quad (2)$$

and

$$\eta = \frac{1}{5} \frac{n_e p_F^2}{m_R} \tau_\eta, \quad (3)$$

where n_e is the electron number density, p_F is the electron Fermi momentum, and m_R is the free electron effective mass, allowing for the effects of relativity, which is given by

$$m_R = \frac{p_F}{v_F} = \left[m_e^2 + \left(\frac{p_F}{c} \right)^2 \right]^{1/2}. \quad (4)$$

In equation (2) k_B is the Boltzmann constant, and T is the temperature. Here m_e is the electron rest mass. The characteristic relaxation times calculated in the simplest variational approximation are

$$\tau_\sigma^{-1} = \frac{m_R}{12\pi^3 \hbar^3 n_e} \int_0^{2k_F} k^3 dk |v(k, 0)|^2 \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \beta \hbar \omega n(\omega) S(k, \omega) \quad (5)$$

$$\tau_\kappa^{-1} = \frac{m_R}{12\pi^3 \hbar^3 n_e} \int_0^{2k_F} dk |v(k, 0)|^2 \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \beta \hbar \omega n(\omega) S(k, \omega) \left[k^3 + \left(\frac{\beta \hbar \omega}{\pi} \right)^2 k \left(3k_F^2 - \frac{k^2}{2} \right) \right] \quad (6)$$

and

$$\tau_\eta^{-1} = \frac{m_R}{12\pi^3 \hbar^3 n_e} \int_0^{2k_F} dk |v(k, 0)|^2 \left(3k^3 - \frac{3k^5}{4k_F^2} \right) \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \beta \hbar \omega n(\omega) S(k, \omega). \quad (7)$$

Here $k_F = p_F/\hbar$ is the electron Fermi wavenumber, $S(k, \omega)$ is the dynamical structure factor of the ions, as a function of wavenumber and frequency, and $v(k, 0)$ is the matrix element for interaction of electrons with a bare ion. If one neglects the finite size of the nuclei one finds

$$|v(k, 0)|^2 = \left(\frac{4\pi Z e^2}{k^2} \right)^2 \left(1 - \frac{\hbar^2 c^2 k^2}{4\epsilon_F^2} \right) \quad (8)$$

on summing over final electron spin states and averaging over initial ones. The quantity $\epsilon_F = m_R c^2$ is the electron Fermi energy (including the rest mass contribution), $n(\omega) = 1/(e^{\beta\hbar\omega} - 1)$ is the Bose factor, $\beta = 1/k_B T$, and Z is the charge of an ion.

In the liquid regime, since the temperature is high compared with both the melting temperature and the Debye temperature, the $(\beta\omega)^2$ term in equation (6) is negligible, and therefore $\tau_K = \tau_\sigma$. Moreover, at these high temperatures $\beta\omega n(\omega) \approx 1$, and therefore the frequency integral may be performed directly, since

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} S(k, \omega) = n_i S(k),$$

where n_i is the number density of ions and $S(k)$ is the static structure factor. We may therefore write

$$\frac{1}{\tau_\sigma} = \frac{1}{\tau_K} = \frac{4Ze^4 m_R \cdot I_1}{3\pi\hbar^3} \quad (9)$$

$$\frac{1}{\tau_\eta} = \frac{4Ze^4 m_R \cdot I_2}{\pi\hbar^3}, \quad (10)$$

where

$$I_1 = \int_0^{2k_F} \frac{dk}{k} \left(1 - \frac{\hbar^2 c^2 k^2}{4\epsilon_F^2}\right) S(k) \quad (11)$$

and

$$I_2 = \int_0^{2k_F} \frac{dk}{k} S(k) \left(1 - \frac{k^2}{4k_F^2}\right) \left(1 - \frac{\hbar^2 c^2 k^2}{4\epsilon_F^2}\right). \quad (12)$$

For the purpose of making numerical computations, we note that

$$m_R \approx m_e \left[1 + 1.018 \left(\frac{\rho_6}{\mu_e}\right)^{2/3}\right]^{1/2}, \quad (13)$$

where $\rho_6 = \rho/10^6 \text{ g cm}^{-3}$, $\mu_e^{-1} = Z/A$ is the number of electrons per nucleon, and A is the mass number of the ion. Thus

$$\sigma \approx 8.630 \times 10^{21} \frac{\rho_6/\mu_e}{Z(1 + 1.018(\rho_6/\mu_e)^{2/3})} \frac{1}{I_1} \text{ s}^{-1} \quad (14)$$

$$\kappa = 2.346 \times 10^{15} \frac{(\rho_6/\mu_e) T_6}{Z[1 + 1.018(\rho_6/\mu_e)^{2/3}]} \cdot \frac{1}{I_1} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1} \quad (15)$$

$$\eta = 1.893 \times 10^6 \frac{(\rho_6/\mu_e)^{5/3}}{Z[1 + 1.018(\rho_6/\mu_e)^{2/3}]} \cdot \frac{1}{I_2} \text{ gm cm}^{-1} \text{ s}^{-1}. \quad (16)$$

3 Numerical results

The static structure factor is a function of k and of the dimensionless interaction parameter $\Gamma = Z^2 e^2 / (r_i k_B T)$, where the ion cell radius r_i is given by $(4\pi/3)r_i^3 n_i = 1$. In our calculation

we employed Galam & Hansen's (1976) results for the static structure factor. These were obtained by fitting analytic expressions for the long-range behaviour of the pair correlation functions onto results for shorter distances calculated from Hansen's (1973) Monte Carlo simulations of the one-component plasma, and then Fourier transforming. The results of the integration may be fitted by the expressions

$$I_1 = \left(\frac{\pi}{3}\right)^{1/2} \ln Z^{1/3} + \frac{2}{3} \ln \left(1.32 + \frac{2.33}{\Gamma^{1/2}}\right) - 0.484 \frac{(\rho_6/\mu_e)^{2/3}}{1 + 1.018 (\rho_6/\mu_e)^{2/3}} \quad (17)$$

and

$$I_2 = \left(\frac{\pi}{3}\right)^{1/2} \ln Z^{1/3} + \frac{2}{3} \ln \left(1.32 + \frac{2.33}{\Gamma^{1/2}}\right) - 0.475 \frac{1 + 2.036 (\rho_6/\mu_e)^{2/3}}{1 + 1.018 (\rho_6/\mu_e)^{2/3}} + 0.267 \frac{(\rho_6/\mu_e)^{2/3}}{1 + 1.018 (\rho_6/\mu_e)^{2/3}} \quad (18)$$

to within 5 per cent for $\Gamma \geq 4$, and for $\rho_6/\mu_e > 10^{-4}$.

The result of YU is

$$I_1 = \ln \left[\left(\frac{2\pi Z}{3} \right)^{1/3} \left(1.5 + \frac{3}{\Gamma} \right)^{1/2} \right] - \frac{1}{2} \frac{(\rho_6/\mu_e)^{2/3}}{[1 + (\rho_6/\mu_e)^{2/3}]} \quad (\text{YU}). \quad (19)$$

Our value for I_1 is typically about 10 per cent less than theirs, and our electrical and thermal conductivities are therefore about 10 per cent greater than their values. By contrast our results are a factor 2–3 smaller than FI's. Our viscosities are about a factor 2 larger than FI's.

At lower densities the effects of electron–electron interactions must be taken into account. One of the effects here is the screening of the electron–ion interaction. Others are the screening of the ion–ion interaction, vertex corrections to the electron–ion interaction, and changes in the effective mass. To take this into account in the calculation of the transport co-efficients one multiplies the matrix element $v(k, 0)$ in the expression for the relaxation time by $\epsilon^{-1}(k, 0)$ where ϵ^{-1} is the static longitudinal dielectric function. In the random phase approximation the dielectric function for a relativistic electron gas is given by (Jancovici, 1962)

$$\epsilon(k, 0) = 1 + \frac{k_{\text{FT}}^2}{k^2} \left\{ \frac{2}{3} (1 - q^2 B \sinh^{-1} b) - \frac{q}{2} \left(1 - \frac{1}{3q^2 B^2} \right) \ln \left(\frac{1+q}{1-q} \right) + \frac{q}{3} \left(1 - \frac{1}{2q^2 b^2} \right) A \ln \left(\frac{1+A}{1-A} \right) \right\} \quad (20)$$

where k_{FT} is the Fermi-Thomas wave number (neglecting correlation effects)

$$k_{\text{FT}}^2 = \frac{4}{\pi} \alpha \frac{c}{v_F} k_F^2, \quad (21)$$

and α is the fine structure constant,

$$b = \frac{\hbar k_F}{mc} \approx 1.009 \left(\frac{\rho_6}{\mu_e} \right)^{1/3} \quad (22)$$

$$q = \frac{k}{2k_F}, \quad (23)$$

$$A^2 = \frac{(1 + q^2 b^2)}{q(1 + b^2)} \quad (24)$$

and

$$B^2 = \frac{b^2}{1 + b^2}. \quad (25)$$

Screening will affect the relaxation times appreciably when the dielectric function is significantly greater than unity for wavenumbers where the static structure factor has its maximum. This will occur when $k_{FT}^2 Z^{2/3} / k_F^2 \geq 1$ or

$$Z^{2/3} \gtrsim \frac{\hbar v_F}{e^2} = \frac{1.92}{r_s},$$

where

$$r_s = 1.389 \times 10^{-2} \left(\frac{\rho_6}{\mu_e} \right)^{-1/3}$$

is the standard electron density parameter. With the dielectric function included, the integrals are functions not only of the variables Γ and $Z^{1/3}$, as they were in the absence of screening, but also of the dimensionless variable k_{FT}^2 / k_F^2 . In Figs 1–6 we show the integrals I_1 and I_2 for various values of Z and for various values of Γ , as Itoh *et al.* (1983) have also done in recent calculations. We see that the effect of screening is most marked for low Z , and for low densities, as one would expect. For comparison we also show the results given by Yakovlev & Urpin's (1980) calculations.

When the electrons are not treated as a rigid background the interaction between ions will be not a bare Coulomb interaction, but rather a screened one. This will affect the static structure factor of the ions when $k_{FT}^2 Z^{2/3} / k_F^2 \geq 1$. In addition, the effective interaction between ions will contain 3-body and higher order terms. Only variational estimates of the structure factor of a one component plasma with interactions screened by a quantum plasma have been made over a wide range of conditions (see, e.g., Iyetomi, Utsumi & Ichimaru 1981) and it is questionable whether these are accurate enough for estimating transport coefficients. We shall therefore not estimate here how relaxation times are affected by screening of the ion–ion interactions.

To estimate the effect of vertex corrections we may replace ϵ by an effective dielectric function $\tilde{\epsilon}$, which takes into account vertex corrections. Diagrams for $\tilde{\epsilon}$ in the random phase approximation are shown in Fig. 7a, and for $\tilde{\epsilon}$ in Fig. 7b. In the spirit of Hubbard's (1957) approximation (see also Pines & Nozières 1966) we replace the effective exchange interaction in the expression for $\tilde{\epsilon}^{-1}$ by $-4\pi e^2 / (k_0^2 + 2k^2)$. At high k the exchange term therefore cancels half the direct term. We choose k_0 so that for $k \rightarrow 0$ the compressibility K agrees with the exact result, which for small r_s is given to a good approximation by the Hartree-Fock result (see, e.g. Pines & Nozières 1966)

$$\frac{K}{K_0} \approx 1 - \left(\frac{4}{9\pi} \right)^{1/3} \frac{r_s}{\pi}, \quad (26)$$

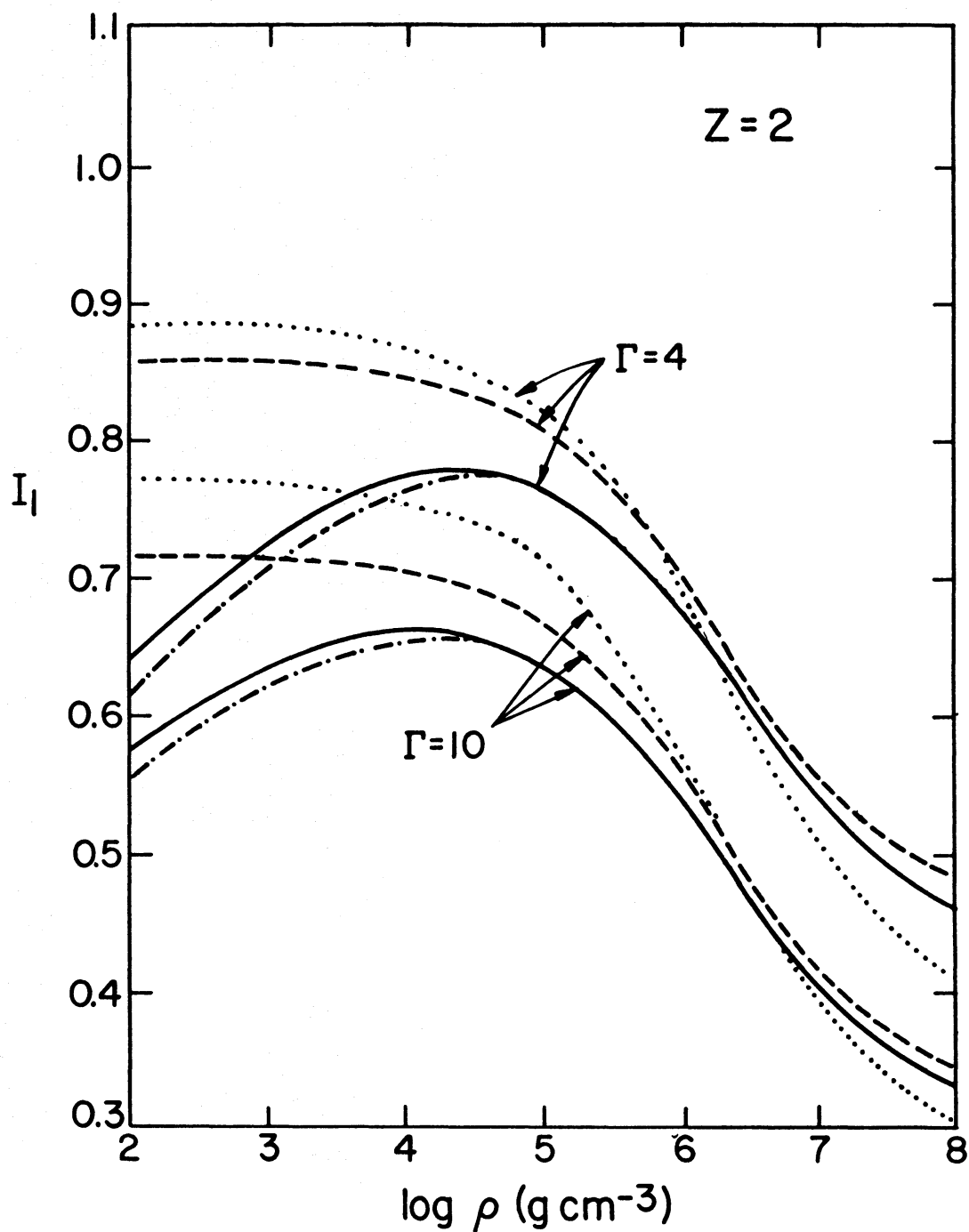


Figure 1. Plot of the integral I_1 versus $\log \rho$ (g cm^{-3}) for $Z = 2$ for $\Gamma = 4$ and $\Gamma = 10$ calculated neglecting screening (dashed line), with screening included in the random phase approximation (dash dotted line) and with the inclusion of vertex corrections (solid line). The results of YU are also shown, as dotted lines.

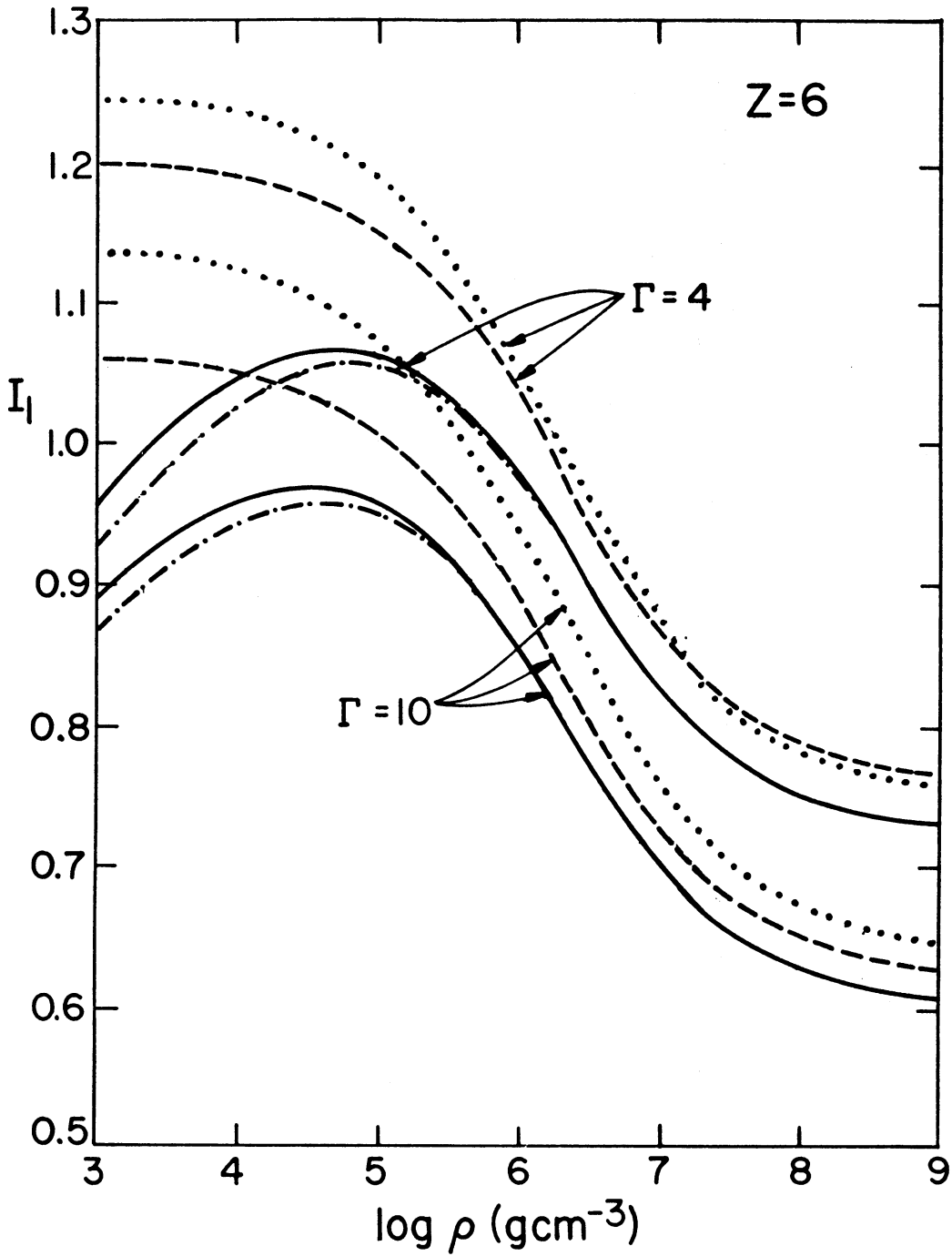


Figure 2. Same as Fig. 1 for $Z = 6$ for $\Gamma = 4$ and $\Gamma = 10$.

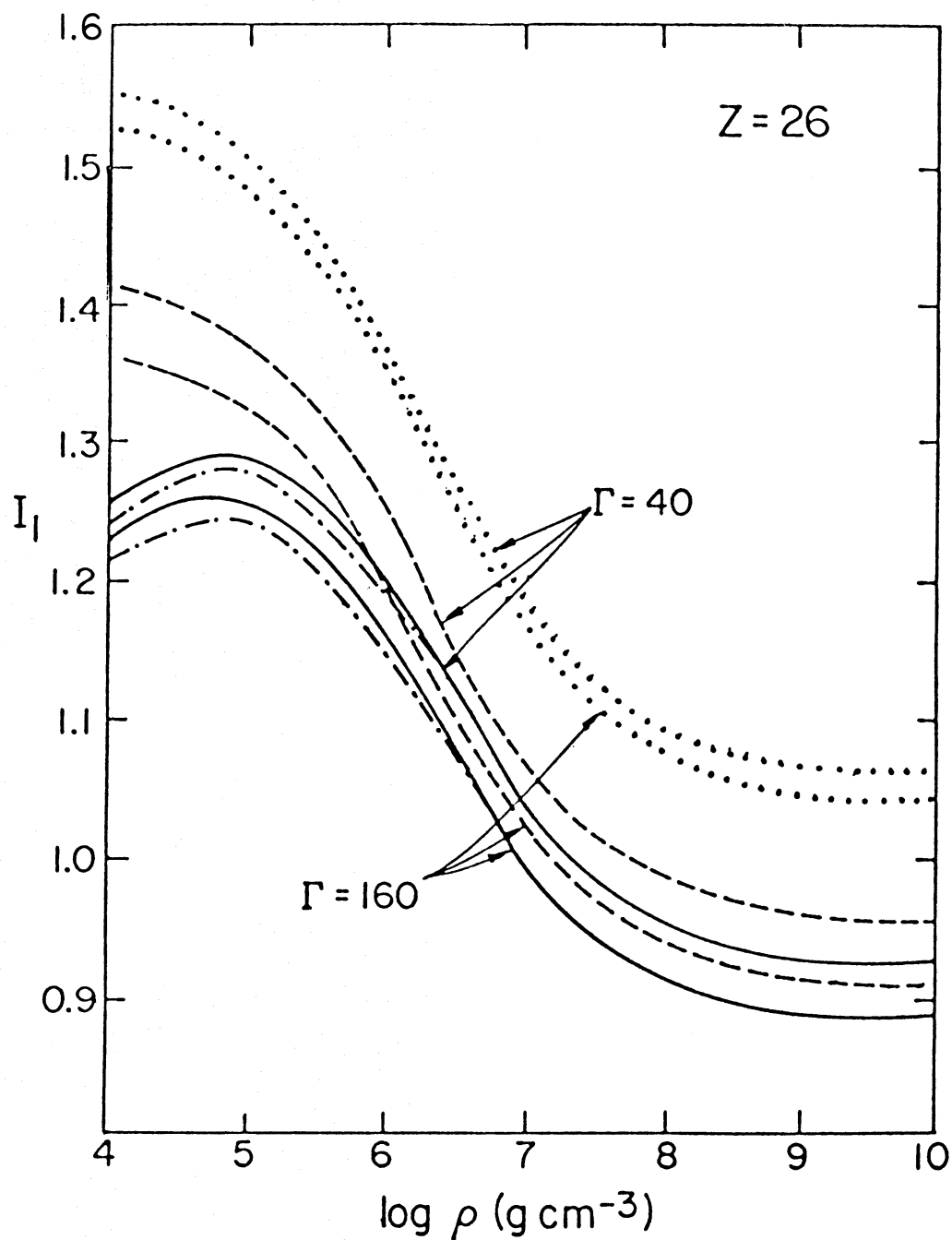


Figure 3. Same as Fig. 1 for $Z = 26$ for $\Gamma = 40$ and $\Gamma = 160$.

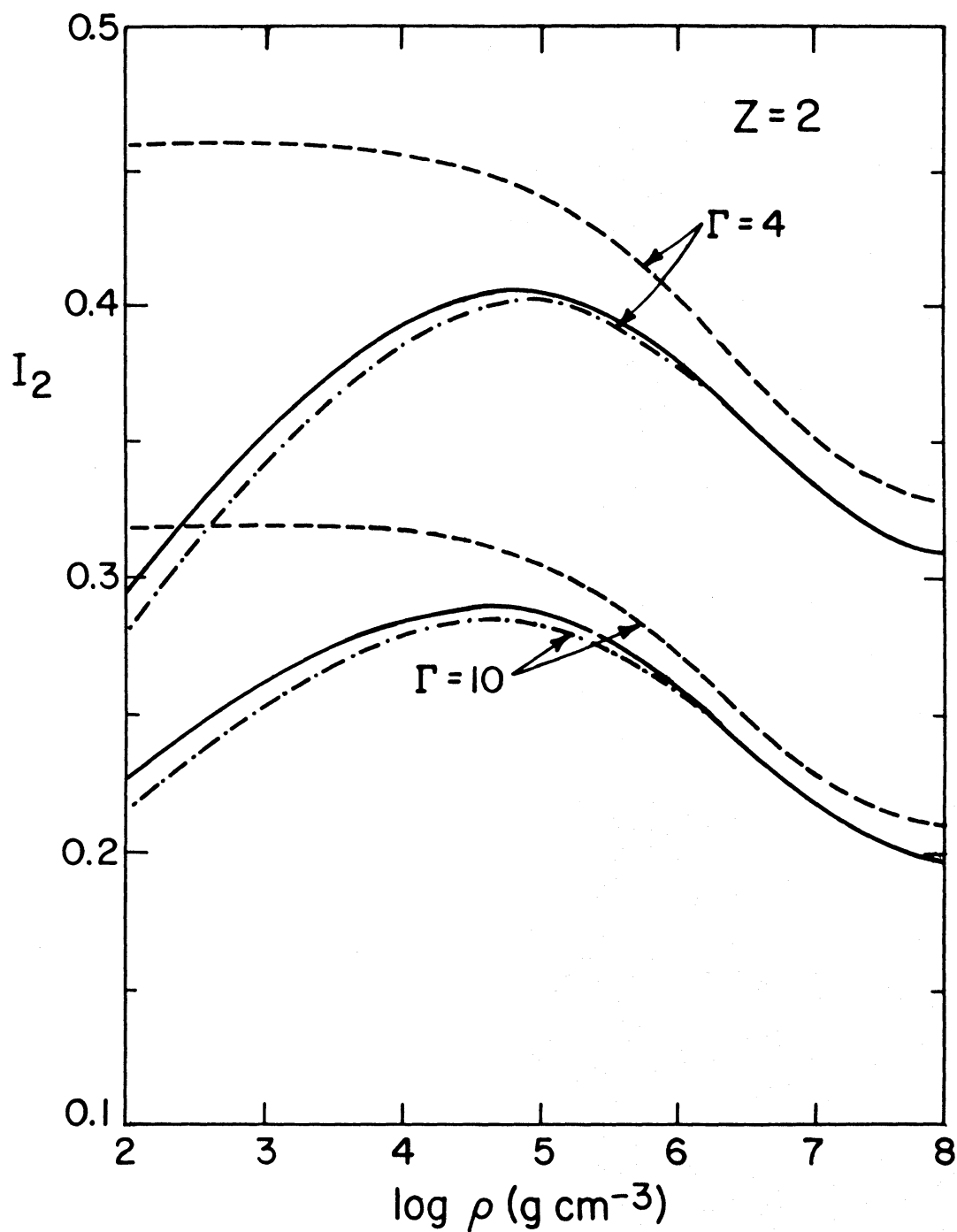


Figure 4. Plot of the integral I_2 versus $\log \rho$ (g cm^{-3}) for $Z = 2$ for $\Gamma = 4$ and $\Gamma = 10$, calculated neglecting screening (dashed line), with screening included in the random phase approximation (dash dotted line) and with the inclusion of vertex corrections (solid line).

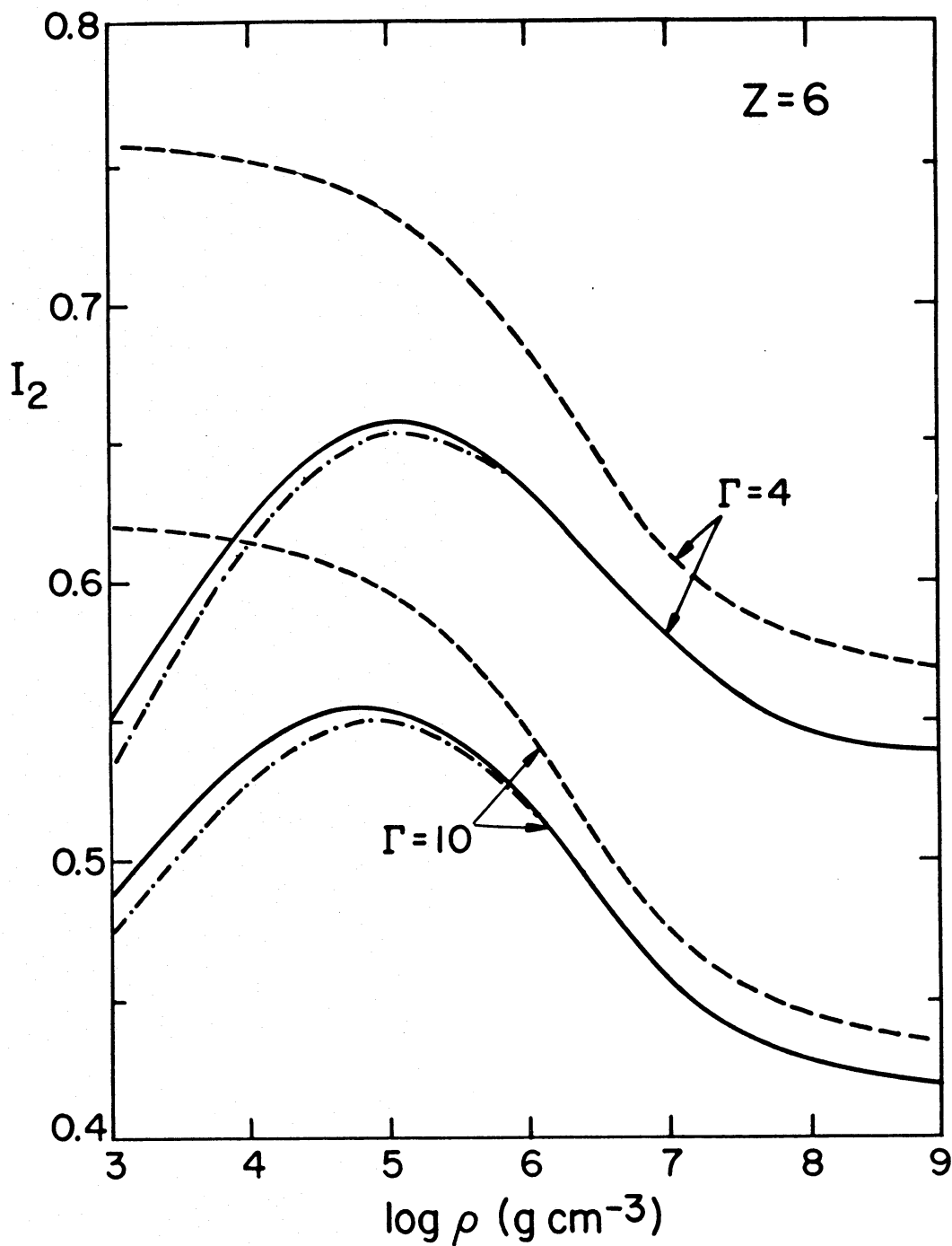


Figure 5. Same as Fig. 4 for $Z = 6$ for $\Gamma = 4$ and $\Gamma = 10$.

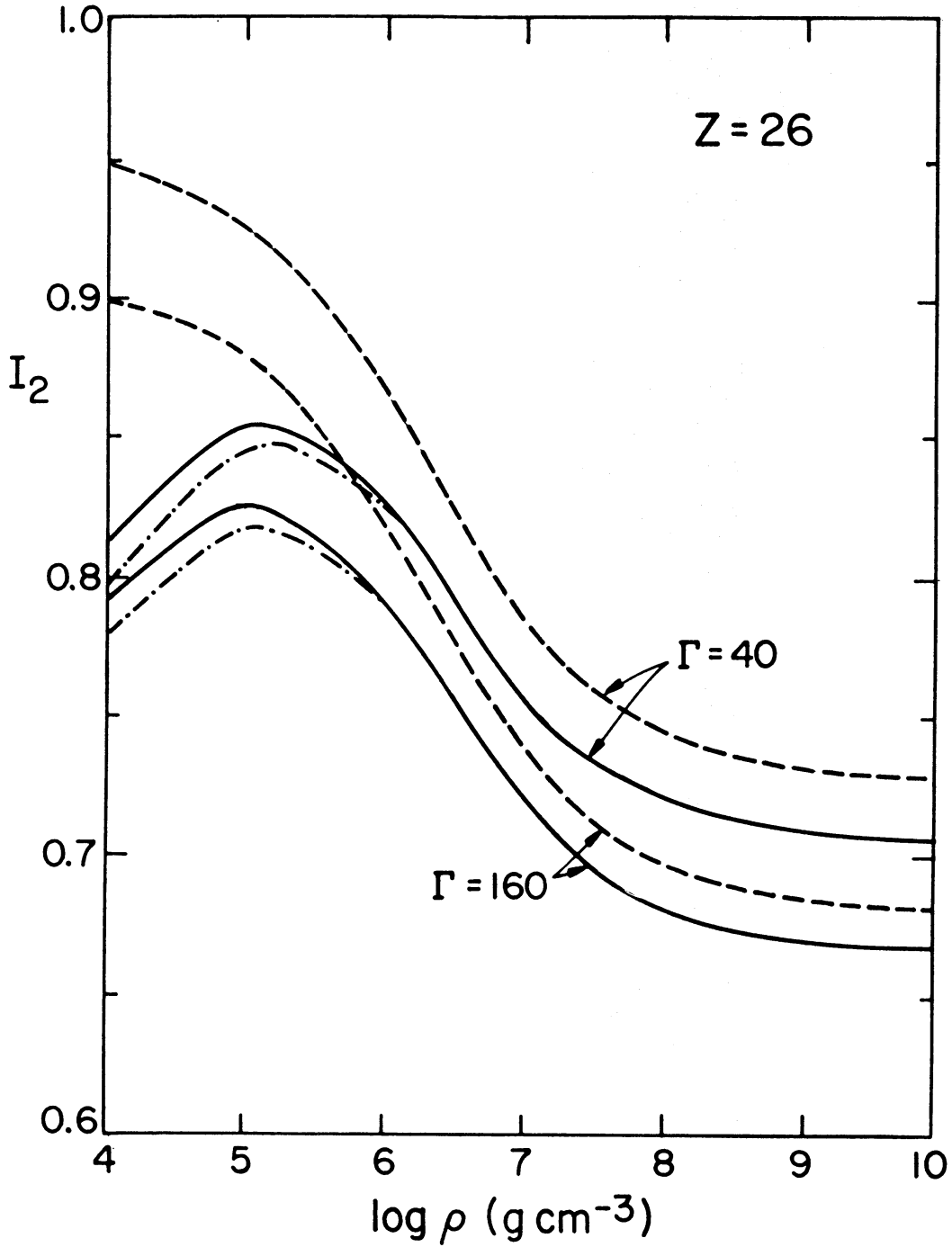


Figure 6. Same as Fig. 4 for $Z = 26$ for $\Gamma = 40$ and $\Gamma = 160$.

$$\begin{aligned}
 \epsilon^{-1}(k, 0) &= 1 + \text{diagram 1} + \text{diagram 2} + \dots \\
 &= 1 - \text{diagram 3}
 \end{aligned} \tag{a}$$

$$\begin{aligned}
 \tilde{\epsilon}^{-1}(k, 0) &= 1 + \text{diagram 4} + \text{diagram 5} + \text{diagram 6} + \dots \\
 &\quad \text{diagram 7} + \dots
 \end{aligned} \tag{b}$$

Figure 7. Diagrams for (a) ϵ the dielectric function in the random phase approximation and (b) the effective dielectric function $\tilde{\epsilon}$ that also takes into account vertex corrections.

where K_0 is the compressibility of the free electron gas. The compressibility expressed in terms of the effective exchange potential is

$$\frac{K}{K_0} = 1 - \frac{4\pi e^2}{k_0^2} N^0(0), \tag{27}$$

where $N^0(0) = m^* p_F / (\pi^2 \hbar^3)$ is the density of states at the Fermi surface for a free electron gas. Therefore one finds

$$k_0^2 = 4k_F^2, \tag{28}$$

a result first derived by Geldart & Vosko (1966). It has previously been used in calculations of transport properties of hydrogen at high densities by Minoo, Deutsch & Hansen (1976). The equality (equation 28) holds for the non-relativistic electron gas, but we shall use it also for the relativistic case, since exchange effects are small then. The result for the effective dielectric function is therefore

$$\begin{aligned}
 \tilde{\epsilon}(k, 0) &= 1 + k_{FT}^2 \left(\frac{1}{k^2} - \frac{1}{2(k^2 + 2k_F^2)} \right) \left\{ \frac{2}{3} (1 - q^2 B \sinh^{-1} b) \right. \\
 &\quad \left. - \frac{q}{2} \left(1 - \frac{1}{3q^2 B^2} \right) \ln \left(\frac{1+q}{1-q} \right) + \frac{q}{3} \left(1 - \frac{1}{2q^2 b^2} \right) A \ln \left(\frac{1+A}{1-A} \right) \right\}.
 \end{aligned} \tag{29}$$

Our estimates of I_1 and I_2 calculated including vertex corrections are also shown in Figs. 1–6. As one can see, they can make differences of up to 4 per cent at the lowest densities.

Finally we estimate the effect on the transport coefficients of the electron–electron contribution to the effective mass. The transport coefficients are given again by equations 1–3 and 5–7, but with m_R replaced by the effective mass, m_R^* , which includes the effect of electron–electron interaction. Thus the transport coefficients scale as m_R^{*-2} . According to Gell-Mann (1957) the effective mass, m^* of a non-relativistic electron gas is given for small r_s by

$$\frac{m^*}{m} = 1 + 0.083 r_s (-\ln r_s - 0.203) \tag{30}$$

and the effects of m^*/m being different from unity are most marked at low densities. For ${}^1\text{H}$ at $\rho = 10^2 \text{ g cm}^{-3}$ m^*/m is 2 per cent above unity, and therefore the transport coefficients are reduced by 4 per cent. Since the vertex correction reduces the transport coefficients by 4 per cent, the total reduction by vertex and effective mass corrections in this extreme case is 8 per cent. At higher densities the effect is less.

4 Conclusion

In summary we have calculated the thermal and electrical conductivities, and the shear viscosity of dense matter under the following assumptions:

- (1) The ions are in the liquid state, which means that Γ must be less than about 160.
- (2) Matter is fully ionized. Thus densities must be very much greater than $10 AZ \text{ gm cm}^{-3}$.
- (3) The electrons are degenerate, which means T must be much less than the Fermi temperature

$$T_F = \frac{\epsilon_F - mc^2}{k_B} \approx 6.04 \times 10^9 \frac{(\rho_6/\mu_e)^{2/3}}{1 + [1 + 1.018(\rho_6/\mu_e)^{2/3}]^{1/2}} \text{ K.} \quad (31)$$

For the static structure factor we use the results of Galam & Hansen (1976). When screening is neglected we find results close to those of Yakovlev & Urpin (1980), but considerably different from those of Flowers & Itoh (1976). When the effects of screening of the electron-ion interaction are taken into account, the characteristic relaxation times increase, but the effect is only 30 per cent even in the most extreme cases, when the density is low and the temperature is high. Note also that these large effects occur only at the limits of validity of our calculations, where either $T \geq T_F/3$ or the matter is not completely ionized.

Calculations of the electrical and thermal conductivities similar to ours, using a more recent result for the static structure factor, but neglecting vertex corrections have recently been carried out by Itoh *et al.* (1983), who also give useful fitting formulae for κ and σ allowing for the effects of screening. These results agree rather well with our calculations without vertex corrections. Vertex and self energy corrections can amount to as much as 8 per cent at low densities.

To improve on the results we presented here, one needs calculations of the static structure factor of a one component plasma interacting via a Coulomb interaction screened by a quantum electron gas.

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